COURSE / AGENT-BASED MODELING

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GROUP MEMBERS

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Intelligent Restaurant Operations

Ontology, knowledge graphs, LLM reasoning, RAG and collaborative adaptive agents

01 SOLUTION OBJECTIVE

A unified restaurant management platform that connects front-of-house, kitchen, inventory and management operations with seven cooperating intelligent agents. Agents use live business data, uploaded operational documents, a shared ontology and knowledge graph to deliver explainable recommendations and controlled automation.

02 COMPLETE PLATFORM CAPABILITIES

CUSTOMER & SALES

POS and order handling: dine-in, takeaway and delivery, customer QR self-ordering, digital menu, order status, tax and service charges, split billing, payments, discounts and receipts.

KITCHEN & DINING FLOOR

Operational execution: kitchen display and preparation tickets, live order queues, table occupancy, seating and waitlist coordination, cleaning workflow and service notifications.

INVENTORY & PROCUREMENT

Supply intelligence: recipe-level ingredient usage, stock movements, reorder policies, waste logging, supplier catalogs, purchasing recommendations and availability control.

PEOPLE, CONTROL & ANALYTICS

Business administration: employee roles, shifts and tasks, workload and performance metrics, sales reporting, menu performance, access permissions, alerts and action audit trails.

03 KNOWLEDGE & REASONING ARCHITECTURE

DATA	SEMANTIC	KNOWLEDGE	REASON	ACT
SQL, uploads, trusted APIs	Extract, align, trace origin	Ontology + linked graph	SQL + Graph + grounded RAG	MAS, approval and feedback

04 SHARED ONTOLOGY, DOCUMENTS & SEMANTIC MAPPING

Ontology and knowledge graph. Define entities such as Restaurant, Order, MenuItem, Ingredient, Recipe, KitchenStation, Employee, Table, Supplier and Policy. Typed relationships include Order contains MenuItem, MenuItem requires Ingredient, Supplier supplies Ingredient and Employee qualifiedFor Station. RDF/OWL captures meaning; SHACL/business rules validate constraints.

Connected intelligence. PostgreSQL provides live orders, quantities and shifts; document upload accepts PDF/DOCX/CSV (recipes, supplier price lists, SOPs, policies). Extraction, entity resolution and semantic mapping align names, units and relationships, with citations, source timestamps, confidence and human review for uncertain mappings.

Seven Collaborative AI Agents

Each agent has a bounded operational responsibility, reads the same governed knowledge context, exchanges structured information with other agents and learns from observed outcomes. A central coordinator resolves conflicting decisions.

01 SMART ORDER COORDINATION

Validate and triage POS/QR orders using menu policy, current stock, kitchen load and customer waiting time.

Outputs: Approve-ready priority proposals, exception flags and feasible alternatives.

Collaboration: Kitchen + Inventory + Supervisor

02 KITCHEN SCHEDULING

Model preparation stations, recipe dependencies, cook capacity and queue age; estimate preparation times and batch compatible dishes.

Outputs: Station assignment, cooking sequence, delay prediction and revised ETAs.

Collaboration: Order + Staff + Inventory

03 PREDICTIVE INVENTORY & PROCUREMENT

Combine sales forecasts with recipe-ingredient links, stock levels, lead times, minimum quantities and supplier documents.

Outputs: Stockout forecasts, availability alerts and supplier-aware purchase drafts.

Collaboration: Order + Kitchen + Sales

04 DYNAMIC STAFF ALLOCATION

Match staff skills and active shifts to waiting tasks, kitchen stations and table sections; monitor fatigue and workload fairness.

Outputs: Task reallocation, coverage alerts and staffing recommendations.

Collaboration: Kitchen + Table + Supervisor

05 TABLE TURNOVER & SEATING

Follow seating, service, billing and cleaning events; incorporate party size, section, capacity and expected table release.

Outputs: Table availability forecasts, cleaning dispatch and seating suggestions.

Collaboration: Order + Staff + Supervisor

06 SALES & MENU OPTIMIZATION

Relate demand, recipe cost, product margin, availability, time-of-day and validated customer preferences.

Outputs: Menu placement, safe upsell combinations, promotion and pricing proposals.

Collaboration: Inventory + Order + Supervisor

07 OPERATIONS SUPERVISOR / ORCHESTRATOR

Purpose: Maintain restaurant-wide objectives and resolve tradeoffs between speed, inventory, staffing, margins and operational policy. Delegate tasks using structured agent-to-agent messages, request evidence, reconcile conflicts and enforce decision limits.

Outputs: Coordinated action plans, escalation alerts, approval requests and traceable justifications backed by data, graph relationships and retrieved policy documents.

SHARED KNOWLEDGE + STRUCTURED A2A

FEEDBACK-DRIVEN, POLICY-CONSTRAINED

From Knowledge to Action

05 ILLUSTRATIVE END-TO-END DECISION

SCENARIO / HIGH-VOLUME BURGER ORDER DURING PEAK SERVICE

01	**Capture**	Order Agent receives an incoming order and checks its menu and service rules.
02	**Cross-check**	Inventory Agent follows recipe links in the graph and detects limited bun stock.
03	**Coordinate**	Kitchen Agent resequences feasible tickets; Staff Agent checks station coverage.
04	**Reason**	Supervisor combines live SQL facts, graph constraints, SOP retrieval and agent evidence.
05	**Execute**	System presents an approved plan, logs actions and observes the real outcome.

06 INTELLIGENCE MECHANISMS & AGENT COMMUNICATION

SEMANTIC DOCUMENT INGESTION

Upload operational PDFs, DOCX and CSV files; extract structured facts and document chunks. Match terminology to the ontology, resolve entities and units, attach provenance and route ambiguous mappings for review.

HYBRID, GROUNDED REASONING

Combine database queries, knowledge-graph traversal, vector RAG and approved external APIs. The LLM interprets context and proposes options; ontology rules, permissions and validators determine what is allowed.

STRUCTURED AGENT-TO-AGENT EXCHANGE

Use typed messages with task ID, sender, objective, evidence links, confidence, constraints and status. Agents can request facts, propose plans, reject infeasible actions and escalate unresolved conflicts.

ADAPTATION & SIMULATION

Record predictions, decisions, approvals and outcomes. Refresh demand and service-time models using measured feedback; test revised policies through an agent-based restaurant digital twin before rollout.

07 RECOMMENDED DELIVERY ARCHITECTURE & CONTROL MODEL

TECHNICAL COMPONENTS

Application: Next.js + TypeScript.
Operational store: PostgreSQL / Supabase.
Documents + retrieval: object storage, pgvector embeddings.
Semantics: RDF/OWL, SHACL and a graph query service.
Agents: event-driven worker, task queue and A2A message contracts.
ABM: Mesa simulation for scenario testing.

SECURITY & GOVERNANCE

Access: role-based permissions and source-level controls.
Evidence: citations, confidence and source freshness.
Safety: deterministic validation; no direct LLM database writes.
Approval: human authorization for purchases, price changes, refunds and order rejection.
Reliability: audit logs, idempotency, retries and fallback rules.

08 SUCCESS MEASURES

Operational KPIs: order acceptance latency, mean/P90 preparation time, stockout rate, ingredient waste, task response time, table turnover, forecast error, accepted recommendations and override rate. Use baseline comparisons and controlled pilots to verify actual impact.